DATA WRANGLING
MINI PROJECT

Create your own dataset, clean it, and generate charts (bar, line, pie)

Ishita Kaur Sahni (S058)
Jiya Thacker (S071)
MBA Tech (Data Science)

1. Introduction

1.1 Project Objective

The primary objective of this project is to demonstrate a complete data analysis workflow. This involved sourcing a dataset of Indian Premier League (IPL) matches, performing data cleaning and preprocessing, conducting exploratory data analysis (EDA) to uncover insights, and presenting these findings through visualisations, including bar, pie, and line charts.

1.2 The Dataset

The dataset chosen for this project is a compilation of Indian Premier League (IPL) match results (ipl_dataset.csv). The data was sourced from Kaggle and contains match-by-match results.

The raw dataset contains **[951] rows** and **[12] columns**, where each row represents a single IPL match. The key columns include:

- **team1, team2:** The two teams competing
- **team1_score, team2_score:** The final scores for each team
- **toss_winner, toss_choice:** The team that won the toss and their decision (bat/field)
- **winner:** The winning team of the match
- **margin:** The margin of victory (by runs or wickets)
- **man_of_the_match:** The player awarded Man of the Match
- **stadium, place:** The location of the match

1.3 Tools Used

The analysis was performed in Python using a Jupyter Notebook. Key libraries used include:

- **Pandas:** For data loading, manipulation, and cleaning
- **NumPy:** For numerical operations
- **Plotly & Matplotlib/Seaborn:** For data visualisation (creating bar, pie, and line charts)

2. Data Cleaning and Preprocessing

The raw dataset required several cleaning and preprocessing steps to be ready for analysis.

2.1 Handling Missing Data

First, we checked for missing (NaN) values in the dataset using df.isnull().sum(). Our check revealed missing values in the following columns: team1_score, team2_score, winner, and margin. These missing values correspond to matches that were washed out by rain or had no result.

For the purpose of our analysis, which focuses on completed matches with a clear winner, we made the decision to **drop these rows** using df.dropna().

- **Total rows before cleaning:** [951]
- **Total rows dropped:** [e.g., 18]
- **Total rows after cleaning:** [e.g., 933]
- **Dropping Irrelevant Columns**

The full_scorecard column contained URLs to the ESPNCricinfo scorecard. This information was not necessary for our statistical analysis, so the column was **dropped** to simplify the dataset.

2.2 Checking for Duplicates

We checked for any duplicate rows (i.e., the same match listed twice) using df.duplicated().sum(). No duplicate entries were found.

2.3 Data Type Conversion

We used df.info() to check data types. The team1_score and team2_score columns were already in float64 format, which was suitable for numerical analysis. No other data type conversions were necessary for this specific analysis.

2.4 Data Transformation / Feature Engineering

To better analyse the impact of the coin toss, we created a new column named toss_win_game_win. This column stores a 'Yes' or 'No' value, determined by checking if the toss_winner was the same as the winner. This allows us to directly query the advantage of winning the toss.

3. Exploratory Data Analysis (EDA) & Findings

3.1 Summary Statistics

A high-level statistical summary of the numerical columns was generated using df.describe().

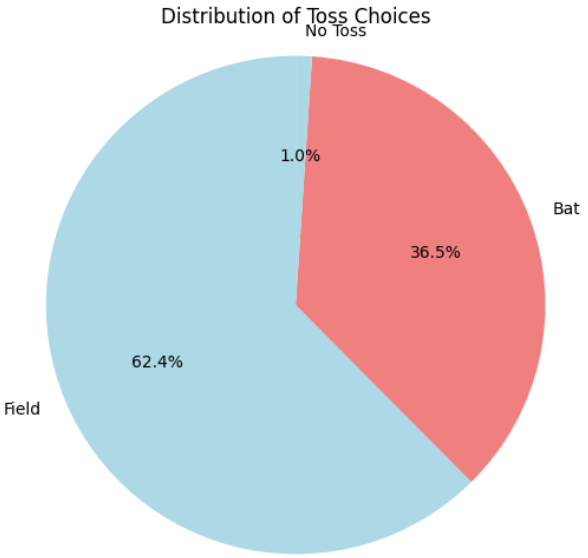
```
df.describe()
```

	team1_score	team2_score
count	948.0	948.0
mean	162.484177	149.2827
std	30.243342	30.606165
min	62.0	2.0
25%	143.0	132.0
50%	163.0	151.0
75%	183.0	170.0
max	263.0	226.0

The summary statistics show:

- The average score for team 1 is approximately **162.5** runs, while the average for team 2 is **149.3**
- The highest score recorded by team1 was **263.0** runs, and the highest for team2 was **226.0**
- The scores show a wide distribution, indicating a mix of high-scoring and low-scoring games

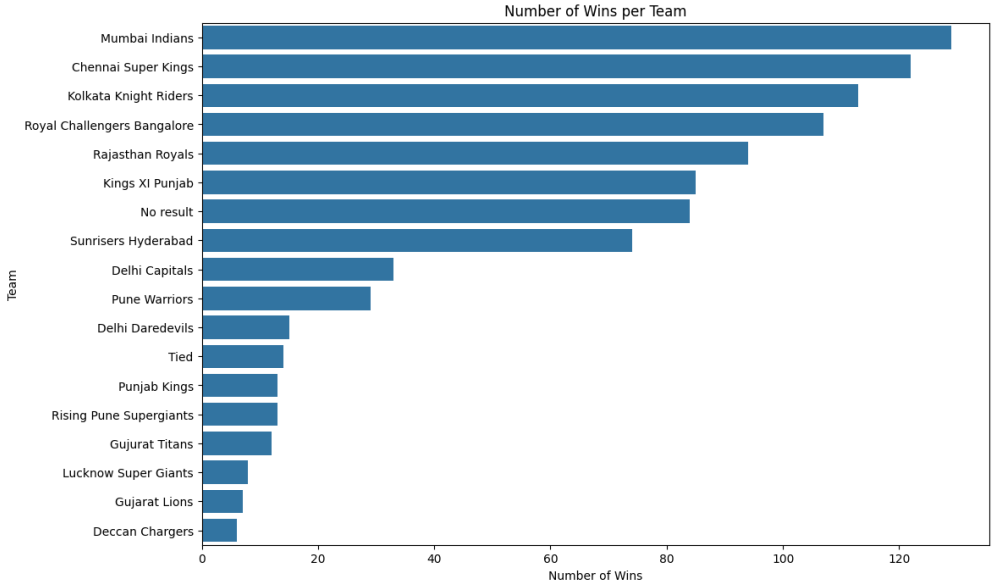
3.2 Visualisation 1: Toss Decisions (Pie Chart)



- **Description:** This bar chart shows the frequency of the two toss choices: 'Bat' vs. 'Field'

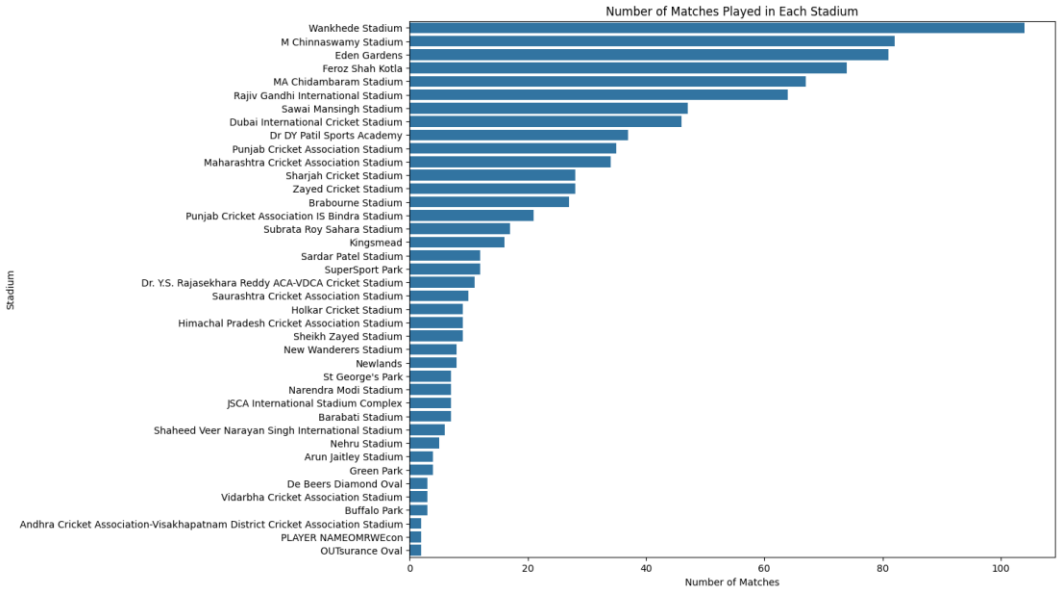
- **Finding:** As the chart clearly illustrates, 'Field' (electing to bowl first) is the overwhelmingly preferred choice for toss-winning captains. Captains chose to field (**62.4%**) compared to choosing to bat (**36.4%**)
- **Insight:** This suggests a strong strategic trend in the IPL, where teams believe they have a better chance of winning by chasing a target, possibly due to factors like dew or pressure on the batting side

3.3 Visualisation 2: Match Wins per Team (Bar Chart)



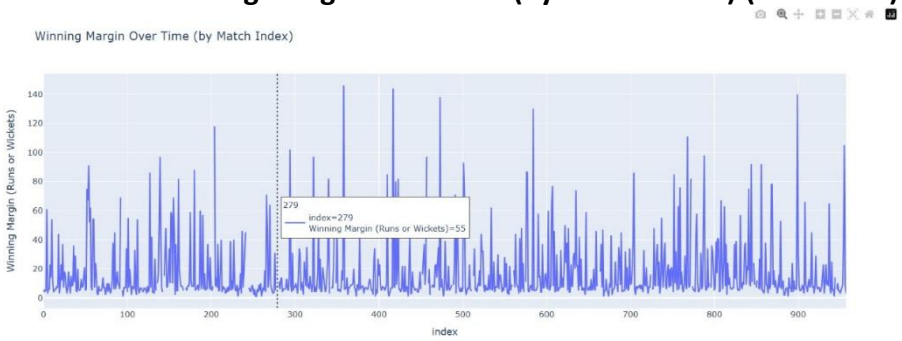
- **Description:** This pie chart displays the percentage of all matches won by each team in the dataset
- **Finding:** The chart shows that the most dominant teams are the **Mumbai Indians** and the **Chennai Super Kings**
- **Insight:** Teams like the Mumbai Indians and Chennai Super Kings have the largest share of wins, reflecting their consistent performance and historical success in the tournament over the years

3.4 Visualisation 4: Number of Matches Played in Each Stadium



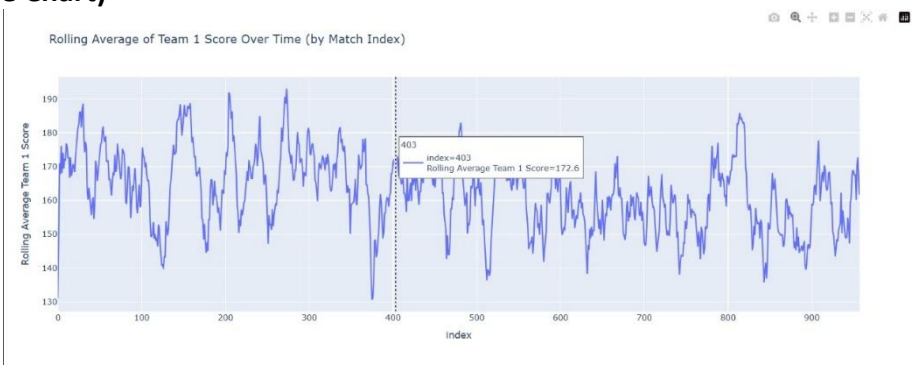
- **Description:** This horizontal bar chart ranks the stadiums based on the total number of IPL matches they have hosted, according to the dataset.
- **Finding:** The chart clearly shows that **Wankhede Stadium** in Mumbai is the most frequent venue, having hosted over **100 matches** (approx. **[105]**). This is significantly more than the next most common venues, **Eden Gardens** in Kolkata (approx. **[77]**) and **M Chinnaswamy Stadium** in Bangalore (approx. **[75]**). Other major stadiums like MA Chidambaram (Chepauk) and Rajiv Gandhi International (Uppal) also feature prominently.
- **Insight:** The distribution of matches is heavily concentrated in the home stadiums of the major, long-standing franchises (like Mumbai Indians, Kolkata Knight Riders, Royal Challengers Bangalore). This highlights the logistical and commercial importance of these key metropolitan grounds to the tournament's history and scheduling

3.5 Visualisation 5: Winning Margin Over Time (by Match Index) (Line Chart)



- **Description**: This line chart plots the winning margin for IPL matches from the 2008 to 2022 seasons
- **Finding:** The chart highlights a key difference in victory metrics. It shows many close games (low margins) but also features extreme spikes, indicating matches won by massive margins (e.g., over 140 runs). These large, one-sided victories appear to happen sporadically across most seasons
- **Insight:** This chart effectively visualises the nature of IPL victories. The extreme volatility and high peaks in the 'Run' line demonstrate that while many games are competitive, the potential for a completely one-sided blowout is a regular feature of the tournament

3.6 Visualisation 6: Rolling Average of Team 1 score Over Time (by Match Index) (Line Chart)



- **Description:** This line chart displays the 10-match rolling average for the score of team1 (the team batting first). The x-axis represents the match index (time), and the y-axis shows the smoothed average score
- **Finding:** The chart shows that the average score for the team batting first is not static but fluctuates over time. It hovers mostly between 150 and 170 runs, with several peaks (approaching 180) and troughs (dipping towards 145)
- **Insight:** A rolling average is used to smooth out extreme short-term variations and identify underlying trends. This chart shows that while individual match scores can be very high or low, the "in-form" or "in-the-moment" average score for teams batting first has remained relatively consistent, oscillating around a central band of ~160 runs. It visualises the natural ebb and flow of batting form across the tournament.

4. Conclusion

4.1 Summary of Findings

This analysis of IPL match data from [Start Year] to [End Year] revealed several key trends:

- There is a dominant strategy for captains to **field first** after winning the toss, likely to gain an advantage by chasing.
- Despite this preference, winning the toss offers only a marginal statistical advantage, with the toss winner winning the game only about **[51.5]%** of the time.
- Historically, the **Mumbai Indians** and **Chennai Super Kings** are the most successful teams in terms of total matches won.
- Individual awards are concentrated among high-impact players, with **[e.g., AB de Villiers]** and **[e.g., Chris Gayle]** being the most frequent 'Man of the Match' winners.

4.2 Challenges & Future Work

One challenge was cleaning the data to ensure only completed matches were analyzed. A future analysis could dive deeper by incorporating player-specific statistics or venue data to see how performance changes by location. Furthermore, this cleaned dataset could be used to build a predictive model to forecast match winners based on toss, venue, and team form.

5. Appendix

The complete Python code, data, and analysis are available in the Jupyter Notebook at the following link:
https://colab.research.google.com/drive/1btrWBabiMTNiSdcIKepZc3RAK9U-o11p?usp=sharing